

D 110211

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Name.....

Reg. No.....

**FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2024**

Mathematics

MTS 5B 08—LINEAR PROGRAMMING

(2020 Admission onwards)

Time : Two Hours

Maximum : 60 Marks

Section A ((Short Answer Type)*All questions can be answered.**Each question carries 2 marks.**Ceiling 20 Marks.*

1. Draw the set of points (x, y) satisfying the constraints

$$2x + y \leq 8, \quad x + 2y \leq 10, \quad x \geq 0, \quad y \geq 0.$$

2. Write the canonical maximization linear programming problem.
3. Define a convex subset of $\mathbb{R}^2$. Also draw a convex set and a non-convex set in $\mathbb{R}^2$.
4. Let S be a convex set in $\mathbb{R}^2$. Define an extreme point of S .
5. Consider the canonical maximum tableau below :

x	y	-1	
1	2	3	$= -t_1$
4	5	6	$= -t_2$
7	8	9	$= f$

State the canonical maximization linear programming problem represented by the tableau above.

6. Write the canonical slack maximization linear programming problem.
7. State Von-Neumann Minimax Theorem.
8. What is complementary slackness of a dual canonical linear programming problem ?

Turn over

9. What is the basic feasible solution of a balanced transportation problem ?
10. Define hyper plane and closed half-space of $\mathbb{R}^n$.
11. What is the mixed strategy of a matrix game ?
12. What is the general balanced assignment problem ?

Section B (Paragraph/Problem Type)

All questions can be answered.

Each question carries 5 marks.

Ceiling 30 marks.

13. Solve graphically : Maximize $f(x, y) = 30x + 50y$ subject to

$$2x + y \leq 8, \quad x + 2y \leq 10, \quad x \geq 0, \quad y \geq 0.$$

14. State Duality Theorem.
15. Solve the transportation problem given below :

7	2	4	10
10	5	9	20
7	3	5	30
20	10	30	

16. Solve the assignment problem given below :

38	21	34
41	14	36
28	20	25

17. Write the simplex algorithm for Maximum Tableau's.
18. Find the von Neumann value and the optimal strategy for each player in the matrix games below :

-1	1	-1	2
-1	-1	1	1
0	1	1	-1

19. What is a two-person zero-sum matrix game ?

Section C (Essay Type)

Answer any **one** of the following questions.

The question carries 10 marks.

20. Solve the canonical linear programming problem using simplex algorithm to the minimum tableau given below :

x_1	20	25	300
x_2	40	20	500
-1	1000	800	0
	$= t_1$	$= t_2$	$= g$

21. Solve the following maximization problem :

Maximize $f(x, y) = x + 3y$ subject to
 $x + 2y \leq 10$, $3x + y \leq 15$, $x \geq 0$, y is unconstrained.

(1 × 10 = 10 marks)